

STAC67H3F Assignment 1: Solutions

Fall 2026

Question 1

The simulation uses an example student number as its seed. Your numbers will differ from these, but the pattern should be the same.

```
set.seed(1004123456)
x <- 1:20
beta0 <- 4; beta1 <- 1.5
Sxx <- sum((x - mean(x))^2)
Sxx
```

```
## [1] 665
```

(a)

```
slope_t <- function(q) {
  y <- beta0 + beta1 * x + rt(length(x), df = q)
  sum((x - mean(x)) * y) / Sxx
}
c(slope_t(5), slope_t(5), slope_t(5))
```

```
## [1] 1.493984 1.537038 1.504299
```

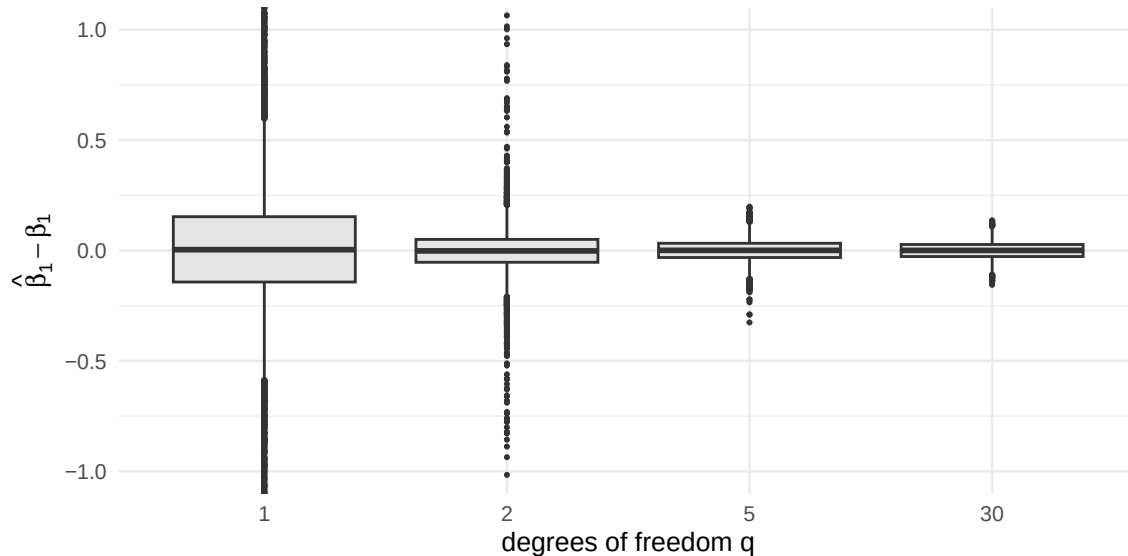
Each call gives a different slope because the errors are redrawn; `x`, `beta0` and `beta1` never change.

(b) The theoretical variance is σ^2/S_{xx} with $\sigma^2 = q/(q-2)$, which exists only for $q > 2$.

```
K <- 10000
qs <- c(1, 2, 5, 30)
sims <- lapply(qs, function(q) replicate(K, slope_t(q)))
tab <- data.frame(
  q = qs,
  mean = sapply(sims, mean),
  sim_MSE = sapply(sims, function(s) mean((s - beta1)^2)),
  theory_Var = ifelse(qs > 2, qs / (qs - 2) / Sxx, NA),
  prop_dev_gt1 = sapply(sims, function(s) mean(abs(s - beta1) > 1)))
options(knitr.kable.NA = "not finite")
knitr::kable(tab, digits = c(0, 4, 4, 5, 4))
```

q	mean	sim_MSE	theory_Var	prop_dev_gt1
1	1.4448	204.0284	not finite	0.0930
2	1.4974	0.0284	not finite	0.0019
5	1.5006	0.0025	0.00251	0.0000
30	1.5004	0.0016	0.00161	0.0000

```
dev <- data.frame(q = factor(rep(qs, each = K)), d = unlist(sims) - beta1)
ggplot(dev, aes(q, d)) +
  geom_boxplot(outlier.size = 0.4, fill = "grey90") +
  coord_cartesian(ylim = c(-1, 1)) +
  labs(x = "degrees of freedom q", y = expression(hat(beta)[1] - beta[1]))
```



(c) For $q = 5$ and $q = 30$ the simulated MSE matches σ^2/S_{xx} and the mean is close to 1.5: the Gauss–Markov setting holds.

For $q = 2$, the errors have mean zero but **infinite variance**. $\hat{\beta}_1$ is still unbiased, since $E(\hat{\beta}_1) = \beta_1 + \sum k_i E(\varepsilon_i) = \beta_1$, but its variance is infinite. The simulated MSE never settles: it is driven by a few enormous values and changes a lot from one run to the next. Gauss–Markov assumes $\text{Var}(\varepsilon_i) = \sigma^2 < \infty$. Without it the theorem says nothing: every linear unbiased estimator has infinite variance, so “smallest variance” is an empty statement.

For $q = 1$ (the Cauchy distribution), the errors **do not even have a mean**, so $E(\hat{\beta}_1)$ does not exist and $\hat{\beta}_1$ cannot be unbiased in any sense. The simulated mean is erratic and meaningless. In the run above it happens to land near 1.5, but that is luck, not unbiasedness: the simulated MSE is in the hundreds, and repeating the simulation with another seed moves the mean by a large and unpredictable amount. The boxplot shows why: most slopes are close to 1.5, and a handful of enormous values drag the average wherever they please. In fact $\hat{\beta}_1 - \beta_1 = \sum_i k_i \varepsilon_i$ has exactly a Cauchy distribution with scale $\sum_i |k_i| = 0.1504$, so a slope more than 1 away from the truth happens with probability $1 - \frac{2}{\pi} \arctan(1/\sum_i |k_i|) = 0.095$, which matches the last column of the table. Both the mean-zero and the finite-variance assumptions fail.

The general lesson: least squares weights every residual by its square, so a handful of extreme errors can dominate the fit. The optimality of least squares is a statement about a class of estimators *under assumptions*, not a guarantee that it behaves well.

Question 2

$E(X) = 1$ and $\text{Var}(X) = (2 - 0)^2/12 = 1/3$; $E(\varepsilon) = 0$, $\text{Var}(\varepsilon) = 4$.

(a) $E(Y) = 3E(X) + E(\varepsilon) = 3$. By independence, $\text{Var}(Y) = 9\text{Var}(X) + \text{Var}(\varepsilon) = 3 + 4 = 7$.

(b) $\text{Cov}(X, Y) = \text{Cov}(X, 3X + \varepsilon) = 3\text{Var}(X) + \text{Cov}(X, \varepsilon) = 1 + 0 = 1$, so

$$\text{Corr}(X, Y) = \frac{1}{\sqrt{1/3}\sqrt{7}} = \sqrt{3/7} \approx 0.6547.$$

(c) $\text{Cov}(\varepsilon, \varepsilon^2) = E(\varepsilon^3) - E(\varepsilon)E(\varepsilon^2) = E(\varepsilon^3) = 0$, because the $N(0, 4)$ density is symmetric about 0, so all its odd moments vanish.

They are **not independent**: ε^2 is a function of ε . For example, condition on $|\varepsilon| < 1$, an event of probability 0.3829. Given that, $\varepsilon^2 < 1$ with certainty, so $P(\varepsilon^2 > 1 \mid |\varepsilon| < 1) = 0$, whereas unconditionally $P(\varepsilon^2 > 1) = P(|\varepsilon| > 1) = 0.6171$. Knowing ε changes what we know about ε^2 . Correlation measures only *linear* association. Zero correlation does not imply independence; here there is a perfect, but non-linear, dependence.

Question 3

(a) Minimise $Q(\beta) = \sum_i (Y_i - \beta x_i)^2$:

$$\frac{dQ}{d\beta} = -2 \sum_i x_i (Y_i - \beta x_i) = 0 \implies b = \frac{\sum_i x_i Y_i}{\sum_i x_i^2}.$$

Since $d^2Q/d\beta^2 = 2 \sum_i x_i^2 > 0$, this is a minimum.

(b) $b = \sum_i c_i Y_i$ with $c_i = x_i / \sum_j x_j^2$, constants because the x_i are fixed. Then

$$E(b) = \sum_i c_i \beta x_i = \beta \frac{\sum_i x_i^2}{\sum_j x_j^2} = \beta,$$

and, since the Y_i are uncorrelated with variance σ^2 ,

$$\text{Var}(b) = \sigma^2 \sum_i c_i^2 = \sigma^2 \frac{\sum_i x_i^2}{(\sum_j x_j^2)^2} = \frac{\sigma^2}{\sum_i x_i^2}.$$

(c) Substituting $b = \sum_j x_j Y_j / \sum_j x_j^2$ directly,

$$\sum_i x_i e_i = \sum_i x_i (Y_i - b x_i) = \sum_i x_i Y_i - b \sum_i x_i^2 = \sum_i x_i Y_i - \frac{\sum_j x_j Y_j}{\sum_j x_j^2} \sum_i x_i^2 = \sum_i x_i Y_i - \sum_j x_j Y_j = 0.$$

Nothing in that calculation forces $\sum_i e_i$ itself to be zero. For a counterexample, take the two observations $(x_1, Y_1) = (1, 1)$ and $(x_2, Y_2) = (2, 1)$. Then

$$b = \frac{(1)(1) + (2)(1)}{1^2 + 2^2} = \frac{3}{5} = 0.6,$$

so $e_1 = 1 - (0.6)(1) = 0.4$ and $e_2 = 1 - (0.6)(2) = -0.2$, giving

$$\sum_i e_i = 0.4 - 0.2 = 0.2 \neq 0.$$

(As a check, $\sum_i x_i e_i = (1)(0.4) + (2)(-0.2) = 0$, as it must be.)

(d) The Y_i are independent $N(\beta x_i, \sigma^2)$, so

$$L(\beta, \sigma^2) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(Y_i - \beta x_i)^2}{2\sigma^2}\right), \quad \ell(\beta, \sigma^2) = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_i (Y_i - \beta x_i)^2.$$

For any fixed σ^2 , maximising ℓ in β means minimising $\sum_i (Y_i - \beta x_i)^2$, so $\hat{\beta}_{\text{MLE}} = b$: **the same as least squares**. Then $\partial \ell / \partial \sigma^2 = -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4} \sum_i (Y_i - \beta x_i)^2 = 0$ gives

$$\hat{\sigma}_{\text{MLE}}^2 = \frac{1}{n} \sum_{i=1}^n (Y_i - \beta x_i)^2.$$

(e) With $E(Y_i) = \beta_0 + \beta x_i$,

$$E(b) = \sum_i c_i(\beta_0 + \beta x_i) = \beta_0 \frac{\sum_i x_i}{\sum_i x_i^2} + \beta, \quad \text{bias}(b) = \beta_0 \frac{\sum_i x_i}{\sum_i x_i^2} = \frac{\beta_0 n \bar{x}}{\sum_i x_i^2}.$$

The bias is zero only if $\beta_0 = 0$ or $\bar{x} = 0$.

(f) Since $\sum_i x_i^2 = S_{xx} + n\bar{x}^2 \geq S_{xx}$,

$$\text{Var}(b) = \frac{\sigma^2}{\sum_i x_i^2} \leq \frac{\sigma^2}{S_{xx}} = \text{Var}(\hat{\beta}_1),$$

with equality only when $\bar{x} = 0$. Dropping the intercept buys precision, but if the true intercept is not zero it costs bias. This is a bias-variance trade-off: leave the intercept out only when there is a good *subject-matter* reason to believe $\beta_0 = 0$, as in the fuel example, not simply because it makes the variance smaller.

Question 4

(a) $\sum_i (x_i - \bar{x})(Y_i - \bar{Y}) = \sum_i x_i Y_i - \bar{Y} \sum_i x_i - \bar{x} \sum_i Y_i + n\bar{x}\bar{Y} = \sum_i x_i Y_i - n\bar{x}\bar{Y} - n\bar{x}\bar{Y} + n\bar{x}\bar{Y} = \sum_i x_i Y_i - n\bar{x}\bar{Y}$.

(b) With $r = S_{xY} / \sqrt{S_{xx} S_{YY}}$, $s_x = \sqrt{S_{xx} / (n-1)}$ and $s_Y = \sqrt{S_{YY} / (n-1)}$,

$$r \frac{s_Y}{s_x} = \frac{S_{xY}}{\sqrt{S_{xx} S_{YY}}} \sqrt{\frac{S_{YY}}{S_{xx}}} = \frac{S_{xY}}{S_{xx}} = \hat{\beta}_1.$$

(c) Using $\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$ and $\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{x}$, the residual is $Y_i - \hat{Y}_i = (Y_i - \bar{Y}) - \hat{\beta}_1 (x_i - \bar{x})$. Squaring and summing,

$$\text{SSE} = \sum_i (Y_i - \bar{Y})^2 - 2\hat{\beta}_1 \sum_i (x_i - \bar{x})(Y_i - \bar{Y}) + \hat{\beta}_1^2 \sum_i (x_i - \bar{x})^2 = S_{YY} - 2\hat{\beta}_1 S_{xY} + \hat{\beta}_1^2 S_{xx}.$$

Since $S_{xY} = \hat{\beta}_1 S_{xx}$, the last two terms combine:

$$\text{SSE} = S_{YY} - 2\hat{\beta}_1^2 S_{xx} + \hat{\beta}_1^2 S_{xx} = S_{YY} - \hat{\beta}_1^2 S_{xx}.$$

(d) From (c) and part (b), $\hat{\beta}_1^2 S_{xx} = r^2 S_{YY}$, so $\text{SSE} = S_{YY}(1 - r^2)$ and $\text{MSE} = S_{YY}(1 - r^2)/(n-2)$. Also from (b), $\hat{\beta}_1 = r \sqrt{S_{YY}/S_{xx}}$. Therefore

$$\frac{\hat{\beta}_1}{\sqrt{\text{MSE}/S_{xx}}} = \frac{r \sqrt{S_{YY}/S_{xx}}}{\sqrt{S_{YY}(1 - r^2)/((n-2)S_{xx})}} = \frac{r \sqrt{n-2}}{\sqrt{1 - r^2}}.$$

Question 5

```
n <- 25; sx <- 10637; sy <- 1869
Sxx5 <- 286432.24; Syy5 <- 5366.56; Sxy5 <- 25056.88
xbar <- sx / n; ybar <- sy / n
b1 <- Sxy5 / Sxx5; b0 <- ybar - b1 * xbar
SSE5 <- Syy5 - Sxy5^2 / Sxx5; MSE5 <- SSE5 / (n - 2)
se_b1 <- sqrt(MSE5 / Sxx5); se_b0 <- sqrt(MSE5 * (1/n + xbar^2 / Sxx5))
ci90 <- b1 + c(-1, 1) * qt(0.95, n - 2) * se_b1
t15 <- (b1 - 0.15) / se_b1; p15 <- 2 * pt(-abs(t15), n - 2)
```

(a) $\bar{x} = 10637/25 = 425.48$ and $\bar{Y} = 1869/25 = 74.76$.

$$\hat{\beta}_1 = \frac{S_{xY}}{S_{xx}} = \frac{25056.88}{286432.24} = 0.08748, \quad \hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{x} = 37.539,$$

so the fitted line is $\hat{Y} = 37.54 + 0.0875x$.

(b) $SSE = S_{YY} - S_{xY}^2/S_{xx} = 3174.6$, so $MSE = SSE/23 = 138.026$ (and $\hat{\sigma} = 11.748$ feet).

$$se(\hat{\beta}_1) = \sqrt{MSE/S_{xx}} = 0.02195, \quad se(\hat{\beta}_0) = \sqrt{MSE \left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} \right)} = 9.631.$$

(c) With $t_{0.95; 23} = 1.7139$:

$$0.08748 \pm 1.7139 \times 0.02195 = (0.0499, 0.1251).$$

We are 90% confident that each additional foot of length is associated with an increase of between 0.05 and 0.125 feet in mean nave height; equivalently, between 5 and 12.5 feet per 100 feet of length.

(d) The claim is $\beta_1 = 0.15$. $H_0 : \beta_1 = 0.15$ against $H_1 : \beta_1 \neq 0.15$.

$$t^* = \frac{\hat{\beta}_1 - 0.15}{se(\hat{\beta}_1)} = \frac{0.08748 - 0.15}{0.02195} = -2.848 \quad \text{on 23 df.}$$

$|t^*| > t_{0.975; 23} = 2.069$, and the p -value is 0.0091. We reject H_0 at the 5% level: the data are not consistent with an increase of 15 feet per 100 feet of length; the estimated increase is about 8.7 feet.

(e) $\hat{\beta}_0 = 37.5$ would be the mean nave height of a cathedral of length zero. It has no meaningful interpretation: a cathedral of zero length does not exist, and $x = 0$ is far outside the observed range of 182 to 611 feet, so this is an extrapolation. It is simply the height at which the fitted line crosses the vertical axis.

Question 6

(a) $E(\tilde{\beta}_1) = \sum_i c_i(\beta_0 + \beta_1 x_i) = \beta_0 \sum_i c_i + \beta_1 \sum_i c_i x_i$. If $\sum_i c_i = 0$ and $\sum_i c_i x_i = 1$, this equals β_1 for all β_0, β_1 . Conversely, if it equals β_1 for all β_0, β_1 : taking $\beta_0 = 1, \beta_1 = 0$ gives $\sum_i c_i = 0$, and then taking $\beta_1 = 1$ gives $\sum_i c_i x_i = 1$.

(b) The weights k_i satisfy $\sum_i k_i = 0$, $\sum_i k_i x_i = 1$ and $\sum_i k_i^2 = 1/S_{xx}$ (Lecture 4). Then

$$\sum_i k_i d_i = \sum_i k_i c_i - \sum_i k_i^2 = \frac{\sum_i (x_i - \bar{x}) c_i}{S_{xx}} - \frac{1}{S_{xx}} = \frac{\sum_i x_i c_i - \bar{x} \sum_i c_i}{S_{xx}} - \frac{1}{S_{xx}} = \frac{1 - 0}{S_{xx}} - \frac{1}{S_{xx}} = 0,$$

using part (a). Since the Y_i are uncorrelated with variance σ^2 ,

$$\text{Var}(\tilde{\beta}_1) = \sigma^2 \sum_i c_i^2 = \sigma^2 \sum_i (k_i + d_i)^2 = \sigma^2 \sum_i k_i^2 + 2\sigma^2 \sum_i k_i d_i + \sigma^2 \sum_i d_i^2 = \text{Var}(\hat{\beta}_1) + \sigma^2 \sum_i d_i^2,$$

which is at least $\text{Var}(\hat{\beta}_1)$, with equality only when every $d_i = 0$, that is, when $\tilde{\beta}_1 = \hat{\beta}_1$.

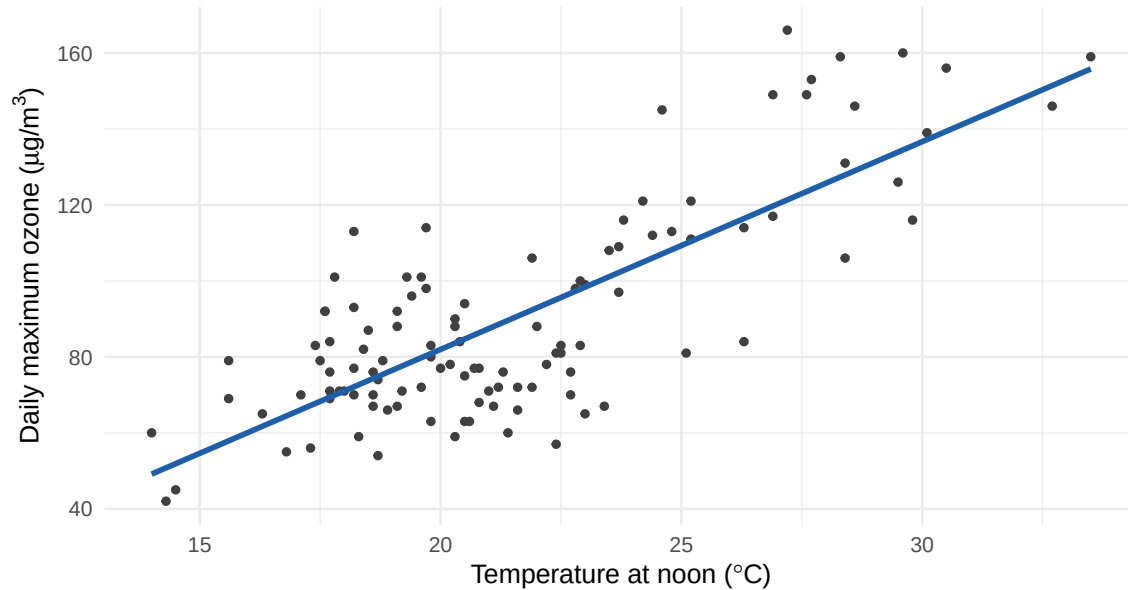
Question 7

```
oz <- read.table("ozone.txt", header = TRUE)
dim(oz)
```

```
## [1] 112 13
```

(a)

```
ggplot(oz, aes(T12, maxO3)) +
  geom_point(size = 1, colour = "grey25") +
  geom_smooth(method = "lm", se = FALSE, colour = "#1F5FA9") +
  labs(x = expression("Temperature at noon (" * degree * "C)"),
       y = expression("Daily maximum ozone (" * mu * "g/m"^3 * ")"))
```



There is a clear positive, roughly linear relationship: warmer days tend to have higher ozone maxima. The scatter around the line is substantial.

(b)

```
xo <- oz$T12; yo <- oz$maxO3; no <- nrow(oz)
Sxx7 <- sum((xo - mean(xo))^2)
Sxy7 <- sum((xo - mean(xo)) * (yo - mean(yo)))
b1o <- Sxy7 / Sxx7
b0o <- mean(yo) - b1o * mean(xo)
fit <- lm(maxO3 ~ T12, data = oz)
rbind(formula = c(b0o, b1o), lm = coef(fit))
```

```
##          (Intercept)      T12
## formula  -27.41964  5.468685
## lm       -27.41964  5.468685
```

$\hat{\beta}_1 = 5.469$: each additional degree of noon temperature is associated with an increase of about $5.47 \mu\text{g}/\text{m}^3$ in the **mean** daily ozone maximum. $\hat{\beta}_0 = -27.42$ would be the mean ozone maximum at 0°C . It is not meaningful: it is negative, which is impossible for a concentration, and 0°C lies far outside the observed range of 14 to 33.5°C .

(c) $H_0 : \beta_1 = 0$ against $H_1 : \beta_1 > 0$.

```
MSE7 <- sum(residuals(fit)^2) / (no - 2)
se1o <- sqrt(MSE7 / Sxx7)
t_obs <- b1o / se1o
p_one <- pt(t_obs, df = no - 2, lower.tail = FALSE)
c(t = t_obs, df = no - 2, p_value = p_one)
```

```
##          t          df      p_value
## 1.325761e+01 1.100000e+02 7.560124e-25
```

```
ci95 <- b1o + c(-1, 1) * qt(0.975, df = no - 2) * se1o
ci95
```

```
## [1] 4.651219 6.286151
```

$t^* = 13.26$ on 110 df, with a one-sided p -value far below 0.01. We reject H_0 at the 1% level: higher noon temperatures are associated with higher ozone maxima. The 95% confidence interval for β_1 is (4.651, 6.286) $\mu\text{g}/\text{m}^3$ per $^\circ\text{C}$.

(d) Under the normal error model, $(\hat{\beta}_1 - \beta_1)/\text{se}(\hat{\beta}_1) \sim t_{n-2}$, so

$$P(|\hat{\beta}_1 - \beta_1| > 0.5) \approx P\left(|T_{110}| > \frac{0.5}{0.4125}\right) = 2P(T_{110} > 1.212) = 0.2281.$$

```
2 * pt(-0.5 / se1o, df = no - 2)
```

```
## [1] 0.2280563
```

It is an approximation because σ is unknown. The exact calculation would use the true standard deviation $\sigma/\sqrt{S_{xx}}$; we used the estimated standard error in its place. Using the normal distribution instead gives 0.2255, almost the same because 110 degrees of freedom is large.